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# Step 1: Import required libraries
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Step 2: Upload an image
uploaded = files.upload()

# Step 3: Read the uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)

# Step 4: Convert the image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 5: Apply Sobel Edge Detection along X-axis
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

# Step 6: Apply Sobel Edge Detection along Y-axis
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Step 7: Convert X and Y results to absolute scale
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Step 8: Combine X and Y edges
sobel_xy = cv2.addWeighted(sobel_x, 0.5, sobel_y, 0.5, 0)

# Step 9: Display the results
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X Axis")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(sobel_xy, cmap="gray")
plt.title("Sobel XY Axis")
plt.axis("off")

plt.show()
```

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every-detail-of-a-sleek-modern-car-captured-in-close-up-photo.jpg(image/jpeg) - 35579 bytes, last modified: 9/1/2026 - 100% done

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Original Image



Sobel X Axis



Sobel XY Axis

